

Membership Function

Features

Core

$$\text{core}_A = \{x \in X \mid \mu_A(x) = 1\}$$

Support

$$\text{supp}_A = \{x \in X \mid \mu_A(x) > 0\}$$

Boundary

$$\text{bound}_A = \{x \in X \mid 0 < \mu_A(x) < 1\}$$

Types of fuzzy sets

Normal fuzzy set

Fuzzy set A with universe of discourse X is a normal fuzzy set if for at least one value $x \in X$, $\mu_A(x) = 1$.

Subnormal fuzzy set

Fuzzy set A is subnormal if it is not normal, i.e. it does not have any value x with membership value 1.

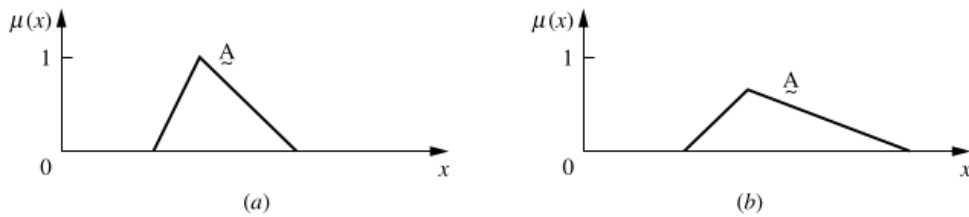


Figure 1: Normal and subnormal fuzzy membership

Convex fuzzy set

Fuzzy set A with universe of discourse X is a convex fuzzy set if its membership value is either monotonically increasing or monotonically decreasing with increasing values of x , i.e. if there are three values $x < y < z$, then both x and z cannot have a membership value greater than that of y at the same time.

$$\mu_A(y) \geq \min(\mu_A(x), \mu_A(z))$$

Concave fuzzy set

Fuzzy set A is concave if it is not convex.

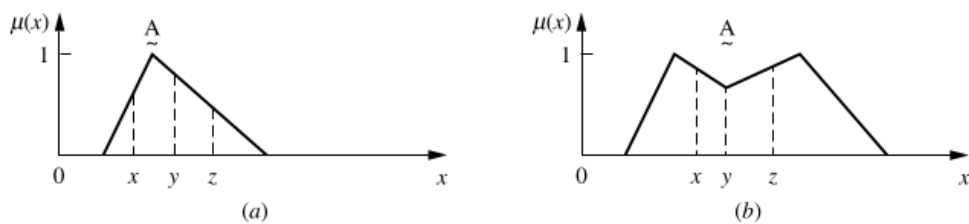


Figure 2: Convex and concave fuzzy membership